It is a disease!

"Later generations will regard *Mengenlehre*[1] as a disease from which one has recovered."

Henry Poincare(1854-1912)[2], *Congress of Mathematicians, Rome, 1908*



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Contributions to topology:

- the foundation of topology for a space of any dimension
- basic concepts and invariants such as Betti numbers and the fundamental group.

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Contributions to topology:

- utilized point-set topology to study Riemann surface ($\rightsquigarrow$ rigorous def. of manifolds)
- topological groups, Lie groups ($\rightsquigarrow$ symmetries in quantum mechanics)

Topology in China



Topology in China



中国拓扑学研究的开始

- 江泽涵(1902-1994): 中国拓扑学的奠基人，1939年开设“形势几何学”课程（西南联大）
- 陈省身(1911-2004): 整体微分几何之父，1946年讲授“拓扑学”（中研院数学所）
- 吴文俊(1919-2017): 著名拓扑学家，科大几何拓扑的源头

B: Some abstract words

Bourbaki: In mathematics we study structure

According to Bourbaki[3], in mathematics we study

1. structures defined on sets,
2. maps that preserve structures.

[3] Bourbaki is the pseudonym of a group of famous mathematicians, including Henri Cartan, Claude Chevalley, Jean Dieudonne, Jean-Pierre Serre, Alexandre Grothendieck, Andre Weil and many others. Bourbaki is most famous for its rigorous presentation of the series of books *Éléments de mathématique* and for introducing the notion of *structures* as the root of mathematics.

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Roughly speaking, a structure is a set endowed with some additional feathers on the set, usually prescribed via subsets (of subsets of ...).

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Mother structures

There are many mathematical structures, among which the following three structures are most elementary (called *mother structures*):

- algebraic structure
- topological structure
- order structure

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(There are also many other structures: metric structure, measure structure, smooth structure, ...)

Topological structure

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$\rightsquigarrow$ As a result, with topology at hand, one can talk about the conception of *continuity* for maps defined on such sets.

Topology v.s. analysis

• So topological structures lie in the center of analysis: it is the topology of $\mathbb{R}$ (or $\mathbb{R}^n$) that allows us to talk about the continuity of (multi-variable) functions.

[4]Warning: To generalize a theorem $\neq$ to write down the theorem in an abstract setting. Usually you will gain in two aspects: first the new theorem should have more applications, and second the abstract form should be able to help you to understand the nature of the original theorem itself.

Topology v.s. analysis

• So topological structures lie in the center of analysis: it is the topology of $\mathbb{R}$ (or $\mathbb{R}^n$) that allows us to talk about the continuity of (multi-variable) functions.

• Many conceptions and theorems we learned in mathematical analysis are topological, and one of the main goals of the first half of this course (which is usually called general topology or point-set topology) is trying to extend these conceptions and theorems to more general spaces. [4]

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Topology v.s. geometry

• Topological structures are also most important in geometry: they tell us how points cluster together in a space, without introducing further structures like distance. In particular, you can change the shape by stretching or twisting or bending. Sometimes topology is called "rubber-sheet geometry"[5]

[5]We will not use such name because it is in fact very misleading.

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• In the second half of this course we will focus on geometric properties that are induced by topology (and not metric).

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C: Some concrete pictures

Topology as a branch of mathematics

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Let's explain this sentence by some pictures.